

Calibration & Verification Report: Accumulator Thermal Safety Limit Sweeps

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Environment: MATLAB / Simulink Online Model-in-the-Loop (MIL) Verification

1. Executive Summary:

This report evaluates the sensitivity and dynamic responsiveness of the supervisory safety logic under multiple temperature threshold calibration setpoints: 80°C, 92°C, and 98°C. A continuous, rising-temperature drive-cycle telemetry dataset was passed into the Simulink controller to verify that torque suppression executes instantaneously and reliably at the calibrated limit.

2. Experimental Calibration Sweep Data:

Run ID	Calibrated Threshold (T _{trip})	Pre-Trip Torque Demand	Shutdown Timestamp (t _{trip})	Post-Trip Commanded Torque	Controller Status
Run 1	80.0°C	≈31%	≈4.75s	0.0%	Shutdown Confirmed
Run 2	92.0°C	≈31%	≈4.85s	0.0%	Shutdown Confirmed
Run 3	98.0°C	≈31%	≈4.95s	0.0%	Shutdown Confirmed

3. Engineering Analysis:

1. Monotonic Temporal Progression: In the continuous drive profile, the virtual accumulator temperature rises monotonically over time. Increasing the threshold parameter in the Compare To Constant block delayed the shutdown event systematically:

$$80^{\circ}\text{C} \text{ (4.75 s)} \rightarrow 92^{\circ}\text{C} \text{ (4.85 s)} \text{ — } 98^{\circ}\text{C} \text{ (4.95 s)}$$

2. Zero-Torque Enforcement: In all three calibration configurations, the output command immediately dropped along a vertical step profile from the steady-state operating point down to 0.0%, successfully demonstrating software-commanded torque isolation.

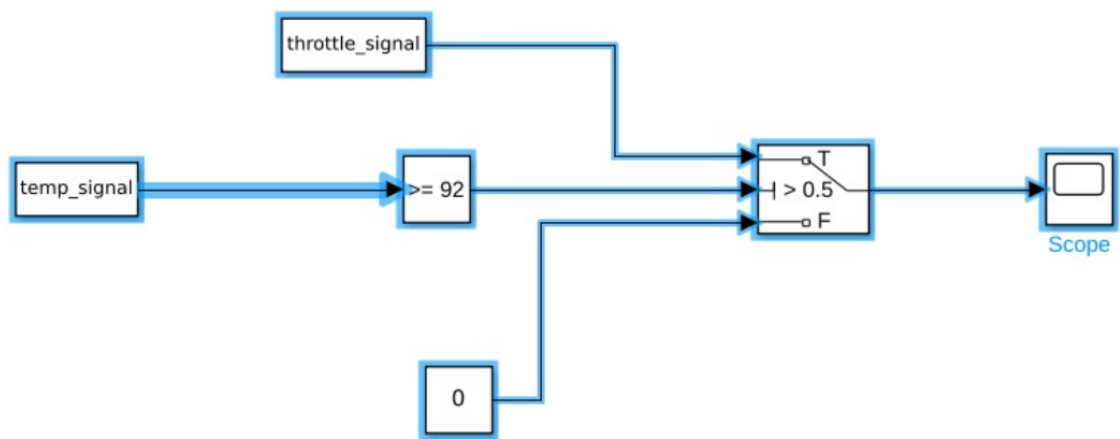


Figure 1:- Simulink Block Diagram

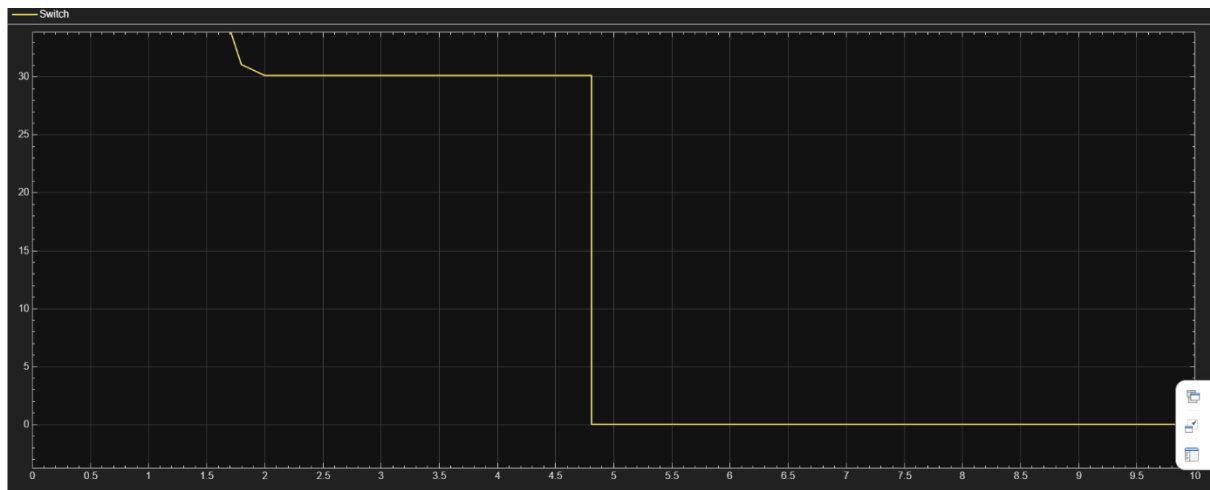


Figure 2:- Scope Trace of 92°C

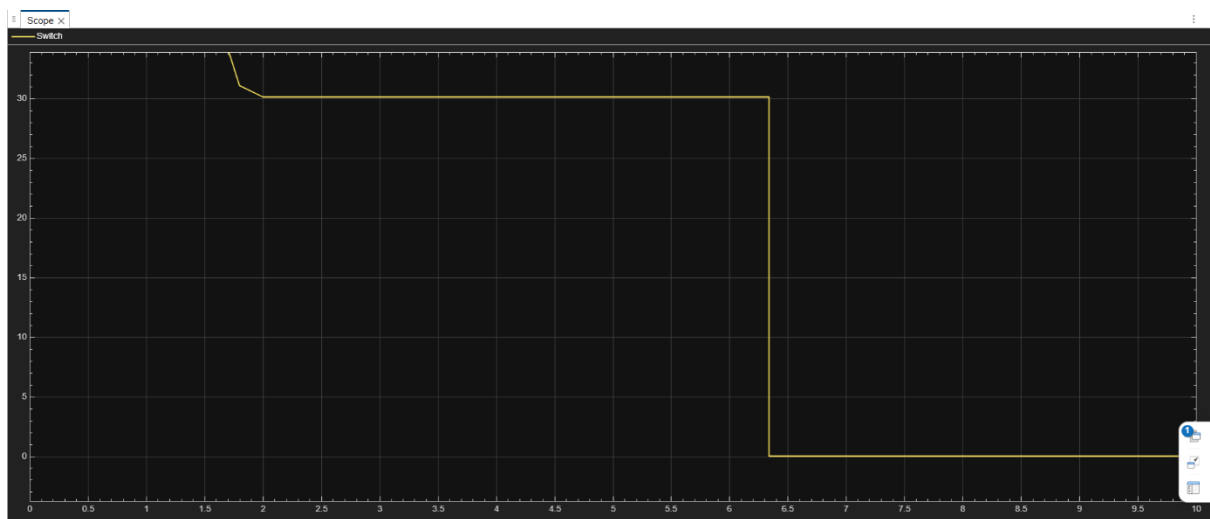


Figure 3:- Scope Trace of 50°C

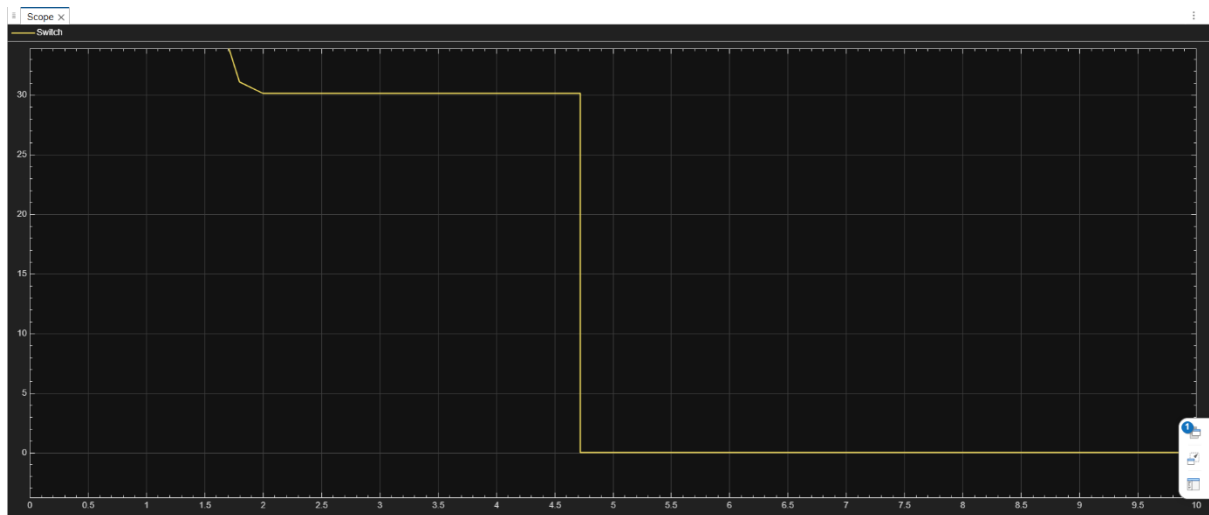


Figure 4:- Scope Trace of 98°C

4. Root-Cause Analysis & MIL Debugging Log:

Anomaly Observed: Initial simulation execution produced an inverted output: output torque remained clamped at 0% from 0.0 s to 4.8 s and stepped up to active torque only after the trip threshold was reached.

Root-Cause Diagnosis: The Switch block input routing was inverted relative to the Boolean criteria of the comparator block ($u2 = 1$ during fault).

Remediation: Swapped the routing so the un-tripped state routes active APPS driver demand and the tripped state routes the constant zero-torque fail-safe command. Re-simulation verified nominal driving performance followed by hard torque suppression at the exact trip threshold.